

AKSHAYKALA SMART DM

AkshayKala Smart DM is an automated Instagram messaging workflow designed to understand incoming customer messages, identify their intent, and deliver an appropriate response automatically.

INTRODUCTION

AkshayKala Junction Pvt Ltd uses Instagram as a customer communication channel, where customers reach out through Direct Messages for product-related and service-related enquiries.

PROBLEM

Instagram DMs are diverse and unpredictable.

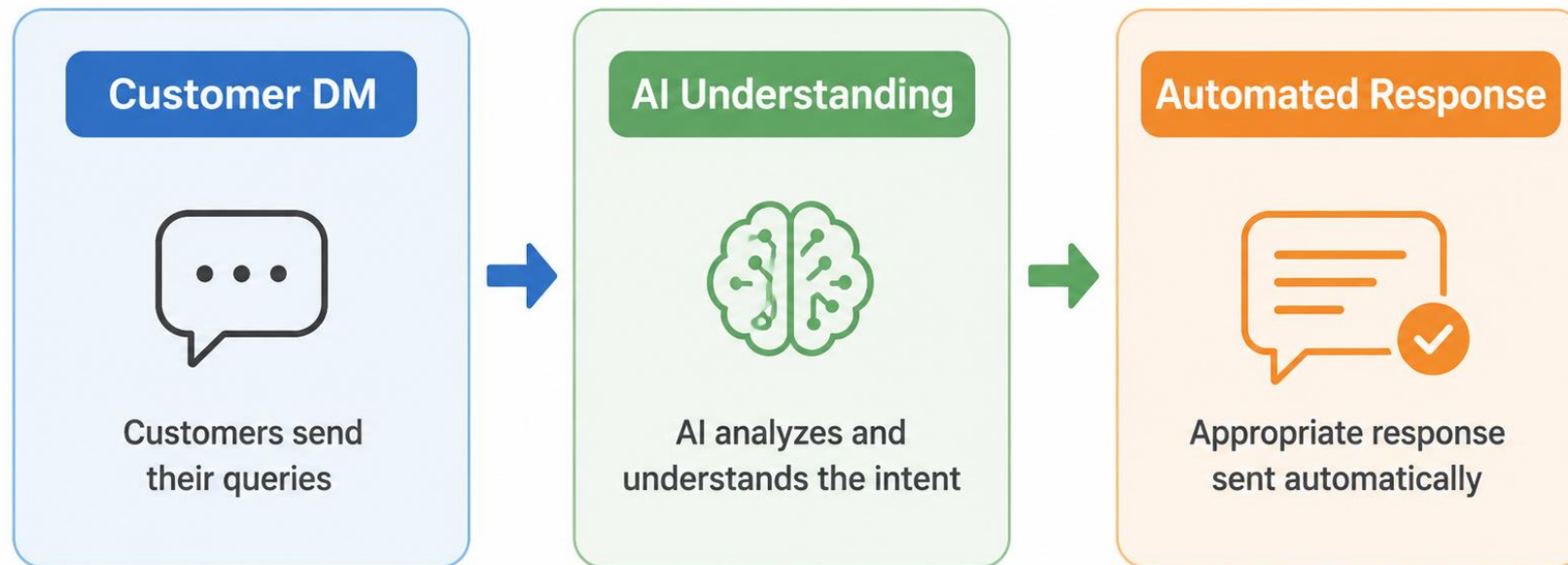
Customers may ask about:

- Products
- Catalogue
- Business enquiries
- Post-purchase support
- General appreciation

Traditional/rule-based automation can struggle with **typos, casual language, Hinglish and different ways of expressing the same intent.**

SOLUTION

An AI-powered automation workflow that understands incoming Instagram messages, identifies the customer's intent, and routes the conversation to the appropriate automated response.



Customer DM

| AI Understanding

| Automated Response

APPROACH

To address the complexity of handling Instagram DMs manually, we implemented the solution using **n8n as the central workflow automation platform**. It allows different services, APIs, processing logic and AI components to be connected into a single automated workflow.

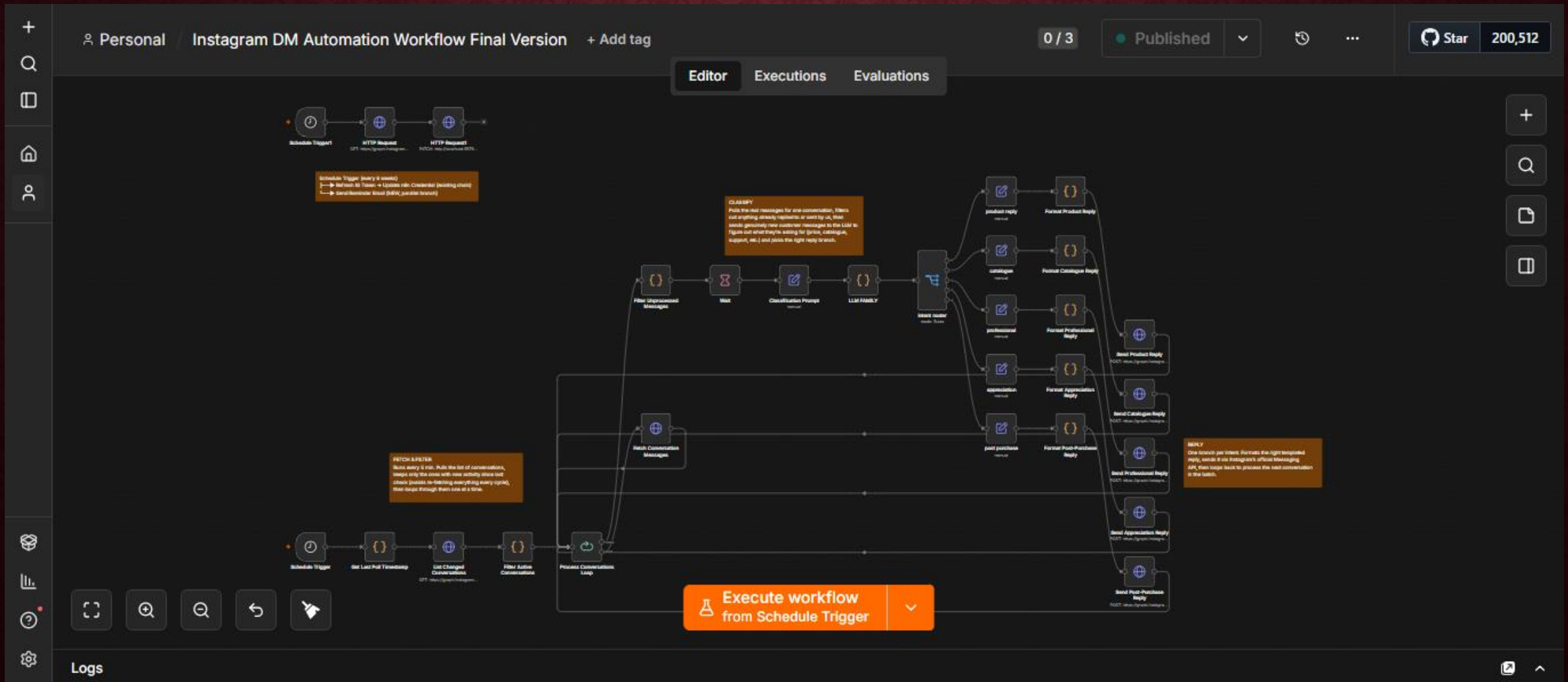
N8N

n8n is a visual, node-based workflow automation platform that enables applications and services to communicate with each other and execute a predefined sequence of operations automatically. Instead of building the entire automation as one large application, the process can be divided into **individual nodes**, where each node performs a specific operation such as:

- Making an API request
- Processing or filtering data
- Applying conditional logic
- Calling an AI/LLM service
- Formatting information
- Sending the final response

This makes complex automation **visual, modular and easier to modify**.

WORKFLOW



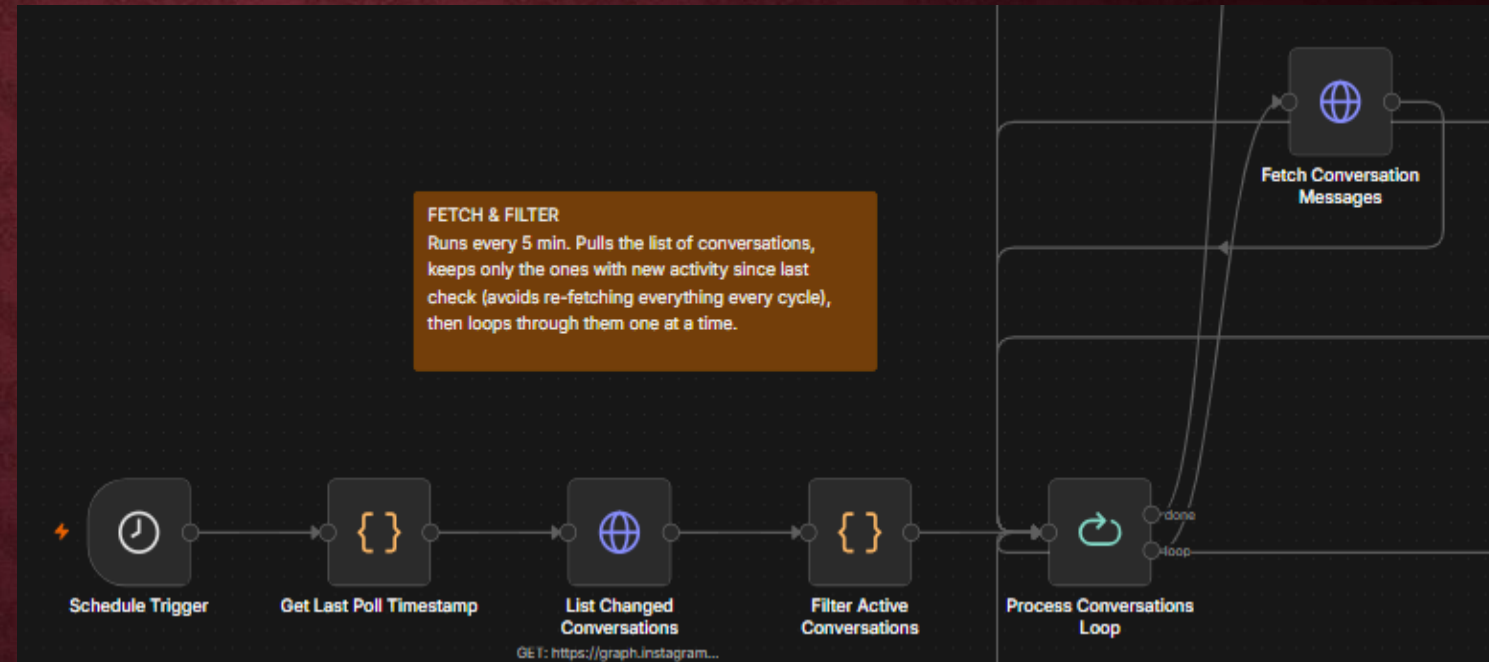
Think of the workflow as a digital receptionist: it receives the message, understands the customer's need, routes it to the right response, and replies automatically.

Batch 1: Fetch & Filter

Batch 1 - Fetch & Filter

- **Schedule Trigger** - Starts the workflow at regular intervals.
- **Get Last Poll Timestamp** - Checks when the workflow last ran.
- **List Changed Conversations** - Fetches recently updated Instagram conversations.
- **Filter Active Conversations** - Keeps only conversations changed since the last check.
- **Process Conversations Loop** - Processes each relevant conversation individually.

Outcome: Only active conversations move to the next stage.

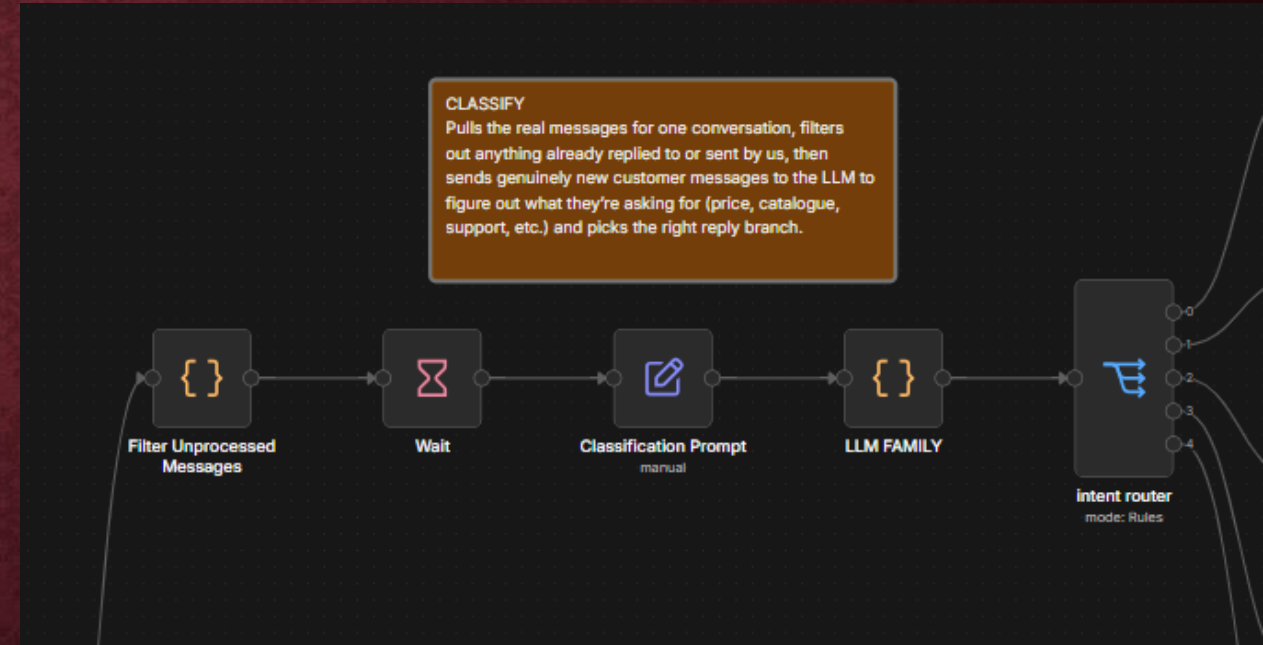


Batch 2: Classify

Batch 2 - Message Processing & Classification

- **Fetch Conversation Messages** - Retrieves messages from the selected conversation.
- **Filter New Messages** - Removes already processed or replied messages.
- **Wait** - Briefly paces the workflow before AI processing.
- **Classification Prompt** - Prepares the customer message for intent classification.
- **LLM Family** - Uses the LLM fallback chain to understand the customer's intent.
- **Intent Router** - Routes the classified message to the appropriate response branch.

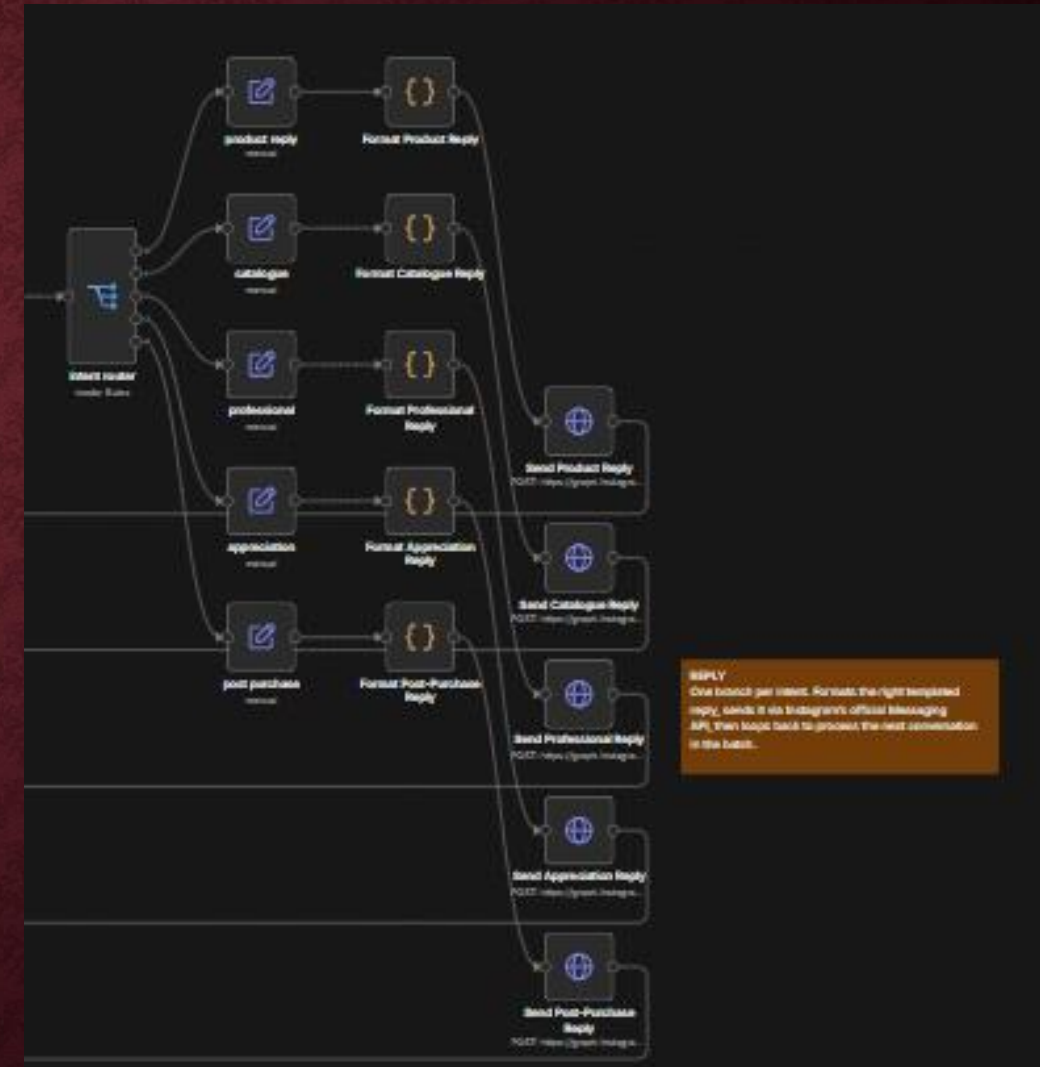
Outcome: The new customer message is **understood** and **assigned to the correct intent**.



Batch 3: Reply

Batch 3 — Intent-Based Response

- **Product Query** — Handles product-related enquiries.
 - **Catalogue** — Responds to catalogue requests.
 - **Professional / Business** — Handles business-related enquiries.
 - **Appreciation** — Sends an appropriate response to appreciation messages.
 - **Post-Purchase** — Handles post-purchase customer communication.
 - **Format Reply** — Prepares the response for the selected intent.
 - **Send Reply** — Sends the response back through the Instagram Graph API.
- Outcome:** Each message receives an **intent-specific automated response**.

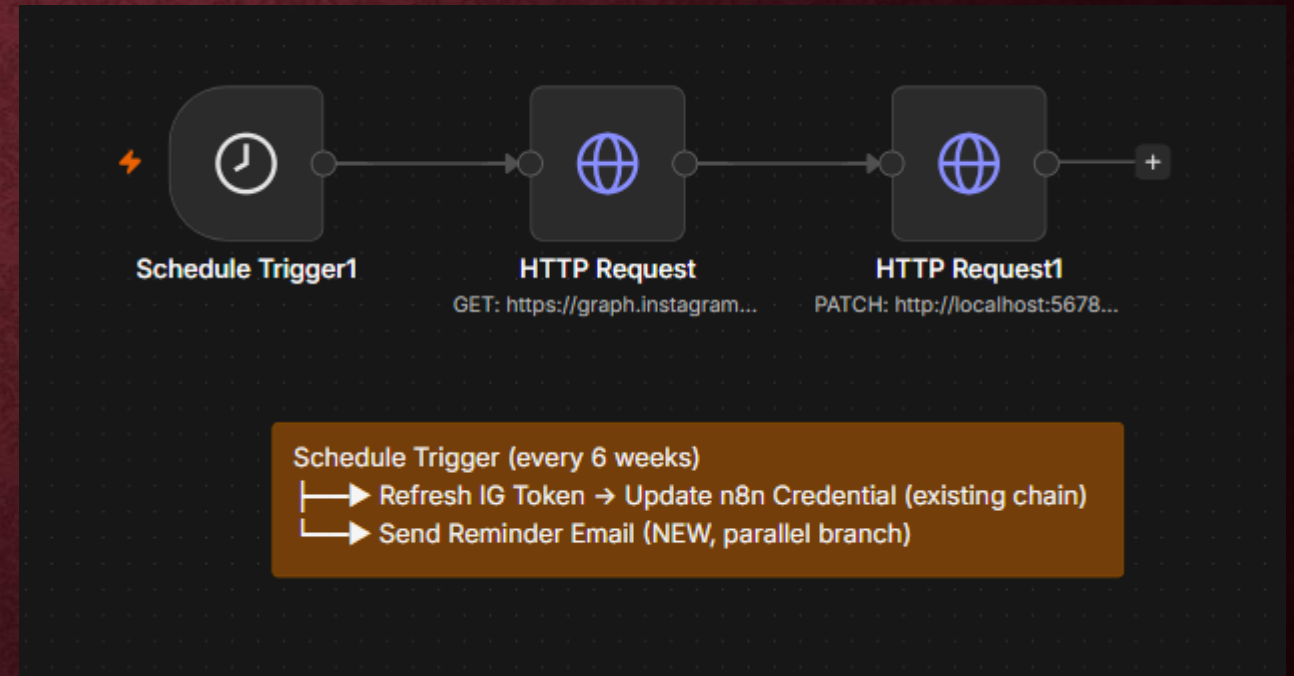


Batch 4: Response Delivery

Batch 4 - Format & Send

- **Format Response** - Prepares the final reply for the selected intent.
- **Response Template** - Applies the appropriate predefined response.
- **Send Reply** - Sends the formatted message to the customer through the Instagram Graph API.
- **Loop Back** - Returns the workflow to the conversation loop to process the next conversation.

Outcome: The customer receives the appropriate automated response, and the workflow continues processing remaining conversations.



Deployment & Hosting

- **Google Cloud VM** - Hosts the production automation environment.
- **Docker** - Runs n8n in an isolated, persistent container.
- **n8n** - Hosts and executes the Instagram DM workflows.
- **Persistent Storage** - Preserves workflows, credentials and execution state.
- **Auto Restart** - Container is configured to restart automatically when required.

The actual deployment is a Google Cloud

e2-micro VM running Ubuntu 22.04, with n8n deployed through Docker and persistent storage configured.

The automation is hosted on a cloud VM, packaged with Docker, and executed through a self-hosted n8n instance.



How the Request Travels

- **Instagram / Meta API** - Provides conversation and message data to the deployed workflow.
- **n8n** - Polls for changed conversations and processes relevant messages.
- **LLM** - Identifies what the customer is asking for.
- **Intent Router** - Sends the message to the appropriate response branch.
- **Response** - The formatted reply is sent back through the Meta Graph API.

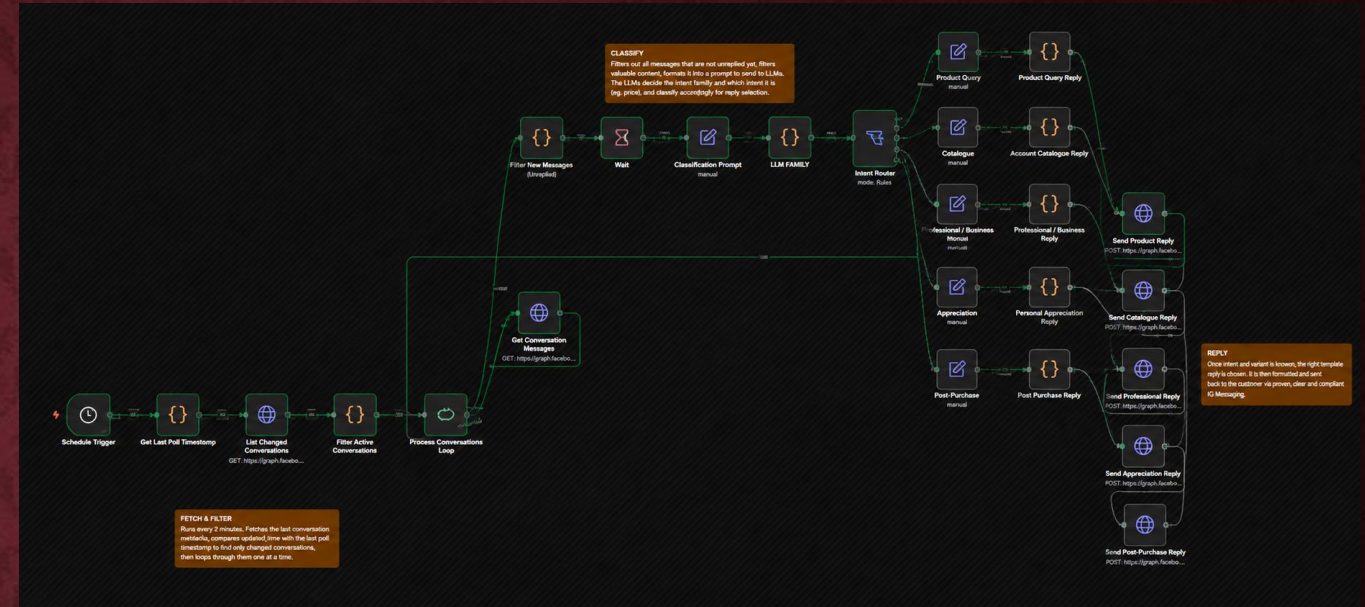


Authentication & Security

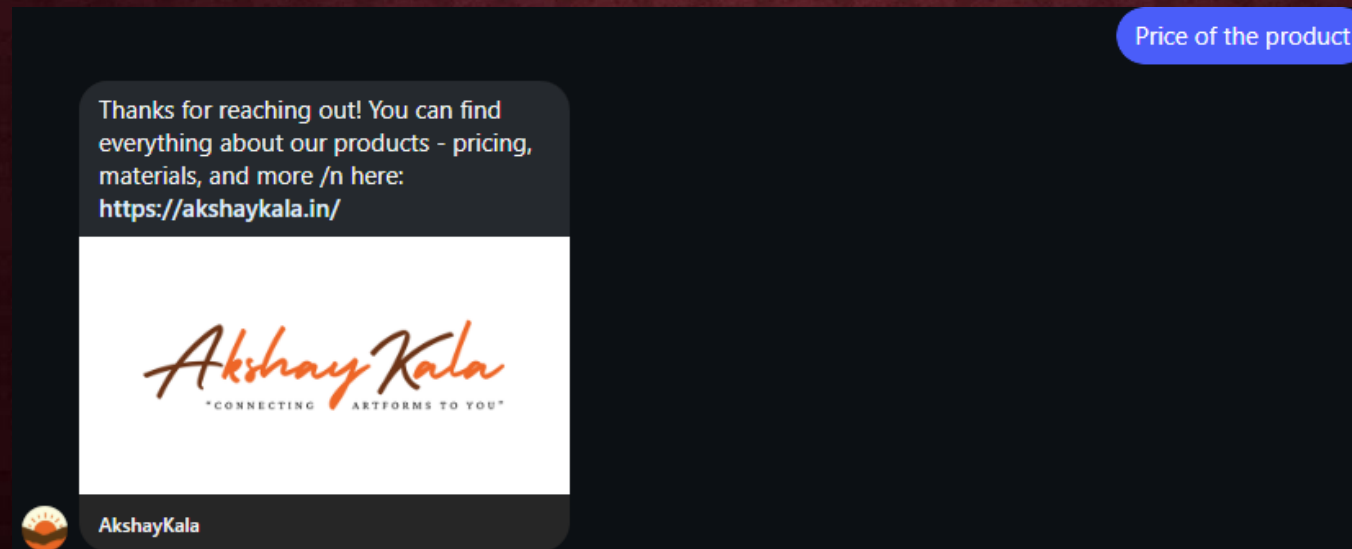
API credentials and LLM keys are managed separately from the workflow logic, while authenticated requests are handled through the deployed n8n environment.

OUTPUT

- Customer sends a DM
- Message intent is identified
- Relevant response branch is triggered
- Automated reply is sent to the customer



Complete image of how the automation is processed



Customer sends a DM and receives an automated response.

Business Impact



Faster Response

Instant automated replies to customer queries.



Reduced Manual Effort

Automates repetitive DM handling and saves valuable time.



Consistent Communication

Ensures accurate and consistent responses for every customer.



Scalable Engagement

Handles a growing volume of conversations effortlessly.



Cost Effective

Reduces operational costs and maximizes resource efficiency.



AkshayKala Smart DM transforms repetitive Instagram DM handling into an automated and scalable customer communication process.

 **Demo Video — Instagram DM Automation Workflow**

Click here to view the complete workflow demonstration

Thank You